

SYDNEY GRAMMAR SCHOOL



2017 Half-Yearly Examination

FORM V

MATHEMATICS 2 UNIT

Tuesday 16th May 2017

General Instructions

- Writing time — 1 hour 30 minutes
- Write using black pen.
- Board-approved calculators and templates may be used.

Total — 80 Marks

- All questions may be attempted.

Section I – 8 Marks

- Questions 1–8 are of equal value.
- Record your answers to the multiple choice on the sheet provided.

Section II – 72 Marks

- Questions 9–14 are of equal value.
- All necessary working should be shown.
- Start each question in a new booklet.

Collection

- Write your name, class and Master on each answer booklet and on your multiple choice answer sheet.
- Hand in the booklets in a single well-ordered pile.
- Hand in a booklet for each question in Section II, even if it has not been attempted.
- If you use a second booklet for a question, place it inside the first.
- Write your name, class and Master on this question paper and hand it in with your answers.
- Place everything inside the answer booklet for Question Nine.

5A: RCF

5B: SO

5C: BR

5D: REJ

5E: LYL

5F: LJF

5G: SDP

5H: CMDB

5P: BDD

5Q: LL

5R: TCW

Checklist

- SGS booklets — 6 per boy
- Multiple choice answer sheet
- Candidature — 190 boys

Examiner

SDP

SECTION I - Multiple Choice

Answers for this section should be recorded on the separate answer sheet handed out with this examination paper.

QUESTION ONE

The expression $\frac{x}{2} + \frac{x}{3}$ simplifies to which of the following?

(A) $\frac{2x}{5}$

(B) $\frac{x^2}{6}$

(C) $\frac{5x}{6}$

(D) $\frac{2x}{6}$

QUESTION TWO

Which of the following is NOT equal to $\sqrt{24}$?

(A) $2\sqrt{6}$

(B) $\sqrt{12} \times \sqrt{2}$

(C) $\frac{1}{2}\sqrt{48}$

(D) $2\sqrt{3} \times \sqrt{2}$

QUESTION THREE

What is the natural domain of the function $f(x) = \frac{1}{x+5}$?

(A) all real values of x , where $x \neq 5$

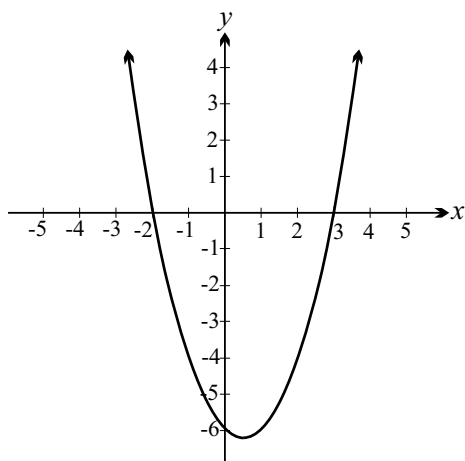
(B) all real values of x , where $x \neq -5$

(C) all real values of x , where $x < 5$

(D) all real values of x , where $x > 5$

Examination continues next page ...

QUESTION FOUR



The above diagram shows the graph of $y = (x - 3)(x + 2)$.

Which of the following is the correct solution to the inequation $(x - 3)(x + 2) < 0$?

- (A) $x > 3$ or $x < -2$
- (B) $x > 2$ or $x < -3$
- (C) $-2 < x < 3$
- (D) $-3 < x < 2$

QUESTION FIVE

Which of the following is the exact value of $\operatorname{cosec}(-60^\circ)$?

- (A) $-\frac{2\sqrt{3}}{3}$
- (B) $\frac{2\sqrt{3}}{3}$
- (C) 2
- (D) -2

QUESTION SIX

What is the perpendicular distance between the point $(3, 5)$ and the line $3x - y + 2 = 0$?

- (A) $\frac{8\sqrt{34}}{17}$
- (B) $\frac{3\sqrt{34}}{17}$
- (C) $\frac{8\sqrt{10}}{5}$
- (D) $\frac{3\sqrt{10}}{5}$

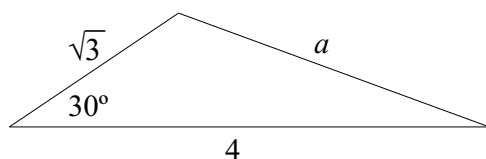
QUESTION SEVEN

Which of the following is the correct factorisation of $54x^3 + 2y^3$?

- (A) $2(3x - y)(9x^2 + 3xy + y^2)$
- (B) $2(3x + y)(9x^2 - 3xy + y^2)$
- (C) $2(3x - y)(3x + 3xy + y)$
- (D) $2(3x + y)(3x - 3xy + y)$

QUESTION EIGHT

Which of the following is the correct value of a ?



- (A) $\sqrt{31}$
- (B) $\sqrt{13}$
- (C) $\sqrt{7}$
- (D) 1

_____ End of Section I _____

SECTION II - Written Response

Answers for this section should be recorded in the booklets provided.

Show all necessary working.

Start a new booklet for each question.

QUESTION NINE (12 marks) Use a separate writing booklet. **Marks**

(a) Evaluate $|2| - |-3 \times 4|$. 1

(b) Factorise fully:

(i) $x^2 + 3x - 10$ 1

(ii) $9x^2 - 16$ 1

(c) Express $\frac{12}{3 - \sqrt{6}}$ as a fraction with a rational denominator. 2

(d) Find the x -intercept of the line $y = 3x - 6$. 1

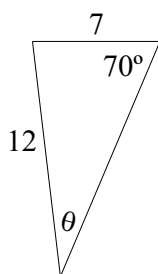
(e) Find the acute angle θ , given that $\cos \theta = \frac{\sqrt{3}}{2}$. 1

(f) Expand and simplify $(x - 3)(x^2 + 3x + 9)$. 1

(g) (i) Find the midpoint M of the interval joining the point $(4, 2)$ and the origin. 1

(ii) Hence find the equation of the line through the point M with gradient 3. 1

(h)



Find the value of θ in the diagram above, correct to the nearest degree. 2

QUESTION TEN (12 marks) Use a separate writing booklet.**Marks**

- (a) Find $f(2)$ when $f(x) = \frac{2}{\sqrt{x^2 + 12}}$. **1**
- (b) (i) Write down the domain of $y = \log_2 x$. **1**
(ii) Write down the range of $y = \log_2 x$. **1**
- (c) Solve the following:
- (i) $3 \leq 2x + 1 \leq 7$ **1**
(ii) $|x - 2| = 10$ **2**
- (d) Find the equation of the line which is parallel to $y = 2x - 5$ and passes through the point $(1, 5)$. Give your answer in general form. **2**
- (e) (i) Complete the square for $f(x) = x^2 + 10x + 17$. **1**
(ii) Hence solve $f(x) = 0$. Leave your answers in exact form. **1**
- (f) Solve $\cos x = \frac{1}{2}$ for $0^\circ \leq x \leq 360^\circ$. **2**

QUESTION ELEVEN (12 marks) Use a separate writing booklet.**Marks**

- (a) Describe the transformation of the circle $x^2 + y^2 = 16$ to the circle $(x + 3)^2 + (y - 2)^2 = 16$. **1**
- (b) Simplify $\sqrt{8} + \sqrt{25} + \sqrt{50}$. **1**
- (c) Simplify $\frac{x^2 + 5x + 6}{2x + 4}$. **2**
- (d) Find the coordinates of the vertex of $y = x^2 - 4x - 32$. **2**
- (e) Solve $-x^2 - 2x + 15 \geq 0$. **2**
- (f) Prove $\frac{\sin x}{\cos x} + \frac{\cos x}{\sin x} = \operatorname{cosec} x \sec x$. **2**
- (g) Determine whether or not the following lines are concurrent: **2**

$$l_1 : y = 2x + 4 \quad l_2 : y = x + 14 \quad l_3 : y = 3x - 6$$

Clearly show your working.

Examination continues next page ...

QUESTION TWELVE (12 marks) Use a separate writing booklet.

Marks

- (a) Solve $x = \frac{1+4x}{x}$. Leave your answer in exact form. **2**
- (b) Find the value of m such that $\frac{2}{\sqrt{3}-\sqrt{2}} = m(\sqrt{2}+\sqrt{3})$. **2**
- (c) (i) Solve $|x-2| < 22$. **2**
(ii) Show your solution on a number line. **1**
- (d) (i) Fully factorise $x^3 - 7x^2 + 10x$. **2**
(ii) Hence solve $x^3 - 7x^2 + 10x = 0$. **1**
(iii) Sketch the curve $y = x^3 - 7x^2 + 10x$, showing any intercepts with the axes. **2**

QUESTION THIRTEEN (12 marks) Use a separate writing booklet.

Marks

- (a) Simplify $\frac{6x-30}{3x^2-75} \div \frac{1}{4x+20}$. **2**
- (b) (i) Sketch $y = \sqrt{9-x^2}$, showing any intercepts with the axes. **2**
(ii) Write down the domain of this function. **1**
(iii) Write down the range of this function. **1**
- (c) In $\triangle ABC$, $\angle BAC = 30^\circ$, $\angle BCA = \theta$, length $AB = 4\sqrt{2}$ cm and length $BC = 4$ cm. **3**
Find the two possible values of θ .
- (d) Find the equation of the perpendicular bisector of the line joining the points $(2, 10)$ and $(8, 0)$. Give your answer in gradient-intercept form. **3**

QUESTION FOURTEEN (12 marks) Use a separate writing booklet.

Marks

- (a) Determine whether $f(x) = x(x^2 - 4)$ is odd, even or neither. **1**
- (b) (i) Sketch $y = 2^x$, showing any intercepts with the axes and any asymptotes. **1**
(ii) Hence sketch $y = -2^x + 2$ and find the x -intercept. **2**
- (c) (i) Accurately sketch $y = 2|x + 2|$ and $y = x + 5$ on the same set of axes. **1**
(ii) Write down the coordinates of any points of intersection. **1**
(iii) Hence or otherwise, solve $2|x + 2| < x + 5$. **1**
- (d) Prove $\operatorname{cosec} x - \sin x = \cot x \cos x$. **2**
- (e) If $\sqrt{15} = ap^2 + b$ and $p = \frac{1}{\sqrt{3}} + \frac{1}{\sqrt{5}}$, find the values of a and b , where a and b are rational numbers. **3**

————— End of Section II —————

END OF EXAMINATION

NAME:

CLASS: MASTER:

SYDNEY GRAMMAR SCHOOL



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FORM V

MATHEMATICS 2 UNIT

Tuesday 16th May 2017

- Record your multiple choice answers by filling in the circle corresponding to your choice for each question.
- Fill in the circle completely.
- Each question has only one correct answer.

Question One

A ☐ B ☐ C ☐ D ☐

Question Two

A ☐ B ☐ C ☐ D ☐

Question Three

A ☐ B ☐ C ☐ D ☐

Question Four

A ☐ B ☐ C ☐ D ☐

Question Five

A ☐ B ☐ C ☐ D ☐

Question Six

A ☐ B ☐ C ☐ D ☐

Question Seven

A ☐ B ☐ C ☐ D ☐

Question Eight

A ☐ B ☐ C ☐ D ☐

Solutions Form V 2 Unit Huls - Yearly 2017

Multiple Choice

$$1) \quad \frac{x}{2} + \frac{x}{3} = \frac{3x}{6} + \frac{2x}{6} \\ = \frac{5x}{6}$$

(C) ✓

$$2) \quad A \rightarrow 2\sqrt{6} = \sqrt{4} \times \sqrt{6} \\ = \sqrt{24} \quad \checkmark$$

$$B \rightarrow \sqrt{12} \times \sqrt{2} = \sqrt{24} \quad \checkmark$$

$$C \rightarrow \frac{1}{2} \sqrt{48} = \sqrt{\frac{1}{4}} \times \sqrt{48} \\ = \sqrt{12} \times$$

(C) ✓

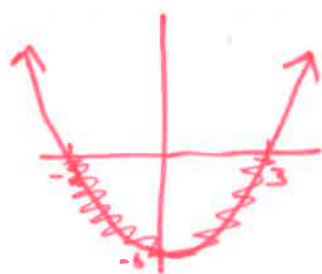
$$D \rightarrow 2\sqrt{3} \times \sqrt{2} = \sqrt{4} \times \sqrt{3} \times \sqrt{2} \\ = \sqrt{24} \quad \checkmark$$

$$3) \quad \text{Denominator cannot equal } 0. \text{ So } x+5 \neq 0 \\ x \neq -5$$

(B) ✓

$$4) \quad (x-3)(x+2) < 0$$

as less than zero, solutions lie under x-axis

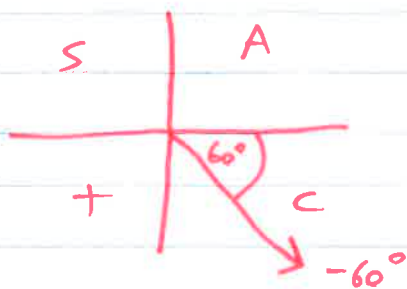


← WANT

$$\text{so } -2 < x < 3$$

(C) ✓

$$5) \quad \operatorname{cosec}(-60) = \frac{1}{\sin(-60)}$$



related angle = 60°

negative sin quadrant

$$\text{so } \sin(-60) = -\sin(60) \\ = -\frac{\sqrt{3}}{2}$$

$$\text{so } \operatorname{cosec}(-60) = \frac{1}{-\frac{\sqrt{3}}{2}}$$

$$= -\frac{2}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}}$$

$$= -\frac{2\sqrt{3}}{3}$$

(A) ✓

$$6) \quad \text{perp. distance} = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

$$x_1 = 3$$

$$a = 3$$

$$y_1 = 5$$

$$b = -1$$

$$c = 2$$

$$= \frac{|3 \times 3 + (-1) \times 5 + 2|}{\sqrt{3^2 + (-1)^2}}$$

$$= \frac{|9 - 5 + 2|}{\sqrt{9 + 1}}$$

$$= \frac{|6|}{\sqrt{10}}$$

$$= \frac{6}{\sqrt{10}} \times \frac{\sqrt{10}}{\sqrt{10}}$$

$$= \frac{6\sqrt{10}}{10}$$

$$= \frac{3\sqrt{10}}{5}$$

(D) ✓

$$\begin{aligned}
 7) \quad 54x^3 + 2y^3 &= 2(27x^3 + y^3) \\
 &= 2((3x)^3 + y^3) \\
 &= 2(3x + y)((3x)^2 - (3x)(y) + y^2) \\
 &= 2(3x + y)(9x^2 - 3xy + y^2) \quad \textcircled{B} \checkmark
 \end{aligned}$$

$$\begin{aligned}
 8) \quad \text{cosine rule} \quad a^2 &= b^2 + c^2 - 2bc \cos A \\
 a^2 &= (\sqrt{3})^2 + 4^2 - 2 \times \sqrt{3} \times 4 \times \cos 30^\circ \\
 &= 19 - 8\sqrt{3} \times \frac{\sqrt{3}}{2} \\
 &= 19 - 12 \\
 a^2 &= 7
 \end{aligned}$$

$$\text{so } a = \sqrt{7}$$

$\textcircled{C} \checkmark$

Question 9

$$\begin{aligned}
 a) \quad |2| - | -3 \times 4 | &= |2| - |-12| \\
 &= 2 - 12 \\
 &= -10 \quad \checkmark
 \end{aligned}$$

$$b) \quad x^2 + 3x - 10 = (x + 5)(x - 2) \quad \checkmark$$

$$ii) \quad 9x^2 - 16 = (3x + 4)(3x - 4) \quad \checkmark$$

$$\begin{aligned}
 c) \quad \frac{12}{3 - \sqrt{61}} \times \frac{3 + \sqrt{61}}{3 + \sqrt{61}} &= \frac{36 + 12\sqrt{61}}{9 - 6} \\
 &= \frac{36 + 12\sqrt{61}}{3} \\
 &= 12 + 4\sqrt{61} \quad \checkmark
 \end{aligned}$$

g d) x-intercept when $y=0$

$$\text{so } 0 = 3x - 6$$

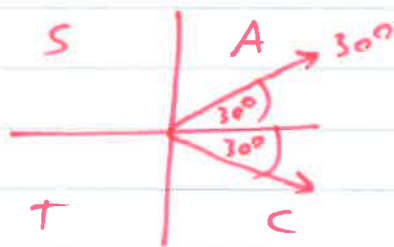
$$3x = 6$$

$$x = 2$$

✓ allow (2, 0)

e) $\cos \theta = \frac{\sqrt{3}}{2}$

related angle $\cos^{-1}\left(\frac{\sqrt{3}}{2}\right) = 30^\circ$



positive and acute angle so $\theta = 30^\circ$ ✓

f) $(x-3)(x^2+3x+9) = x^3-27$ ✓

g i) (4, 2) and (0, 0) $M = \left(\frac{4+0}{2}, \frac{2+0}{2}\right)$

$$M = (2, 1) \quad \checkmark$$

h) $y - y_1 = m(x - x_1)$

$$y - 1 = 3(x - 2)$$

$$y - 1 = 3x - 6$$

$$y = 3x - 5$$

✓ allow $3x - y - 5 = 0$

h) use sin rule $\frac{\sin \theta}{7} = \frac{\sin 70}{12}$ ✓ or similar

$$\sin \theta = \frac{7 \sin 70}{12}$$

$$\theta = \sin^{-1}\left(\frac{7 \sin 70}{12}\right)$$

$$\theta = 33^\circ \quad (\text{nearest degree}) \quad \checkmark$$

Question 10

$$\begin{aligned} \text{a)} \quad f(2) &= \frac{2}{\sqrt{2^2 + 12}} \\ &= \frac{2}{\sqrt{4+12}} \\ &= \frac{2}{\sqrt{16}} \\ &= \frac{2}{4} \\ &= \frac{1}{2} \quad \checkmark \end{aligned}$$

bi) cannot put negatives into a log

so all real values of x , where $x > 0$ ✓

ii) can get any value out of logs

so all real values of y ✓

$$\text{ci)} \quad 3 \leq 2x + 1 \leq 7$$

$$2 \leq 2x \leq 6$$

$$1 \leq x \leq 3 \quad \checkmark$$

iii)

$$|x-2| = 10$$

$$\begin{array}{l} / \quad \backslash \end{array}$$

$$x-2=10$$

$$x=12$$

$$-(x-2)=10$$

$$x-2=-10$$

$$x=-8$$

$$x=12 \quad \checkmark \quad \text{or} \quad x=-8 \quad \checkmark$$

10a)

parallel so gradient = 2 ✓

$$y - 5 = 2(x - 1)$$

$$y - 5 = 2x - 2$$

$$y = 2x + 3$$

$$2x - y + 3 = 0$$
 ✓

xi)

$$f(x) = x^2 + 10x + 17$$

$$= (x + 5)^2 - 25 + 17$$

$$= (x + 5)^2 - 8$$
 ✓

ii)

$$0 = (x + 5)^2 - 8$$

$$8 = (x + 5)^2$$

$$x + 5 = \sqrt{8}$$

or

$$x + 5 = -\sqrt{8}$$

$$x + 5 = 2\sqrt{2}$$

$$x + 5 = -2\sqrt{2}$$

$$x = -5 + 2\sqrt{2}$$

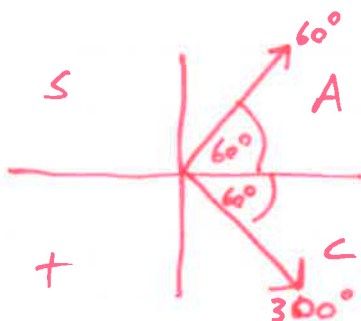
$$x = -5 - 2\sqrt{2}$$
 ✓ both

$$\text{allow } x = -5 \pm \sqrt{8}$$

f)

$$\cos x = \frac{1}{2}$$

$$\text{related angle } x = \cos^{-1}\left(\frac{1}{2}\right) = 60^\circ$$
 ✓



$$\text{so } x = 60^\circ \text{ or } 300^\circ$$
 ✓

Question 11

a) Shift left 3
Shift up 2 ✓ both

b) $\sqrt{8} + \sqrt{25} + \sqrt{50} = 2\sqrt{2} + 5 + 5\sqrt{2}$
 $= 5 + 7\sqrt{2}$ ✓

c) $\frac{x^2 + 5x + 6}{2x + 4} = \frac{(x+2)(x+3)}{2(x+2)}$ ✓ factorise top
 $= \frac{x+3}{2}$ ✓

d) $y = x^2 - 4x - 32$

vertex $x = -\frac{b}{2a}$

$$x = -\frac{(-4)}{2 \times 1}$$

$$x = \frac{4}{2}$$

$$x = 2$$
 ✓

when $x=2$, $y = 2^2 - 4 \times 2 - 32$

$$y = 4 - 8 - 32$$

$$y = -36$$
 ✓

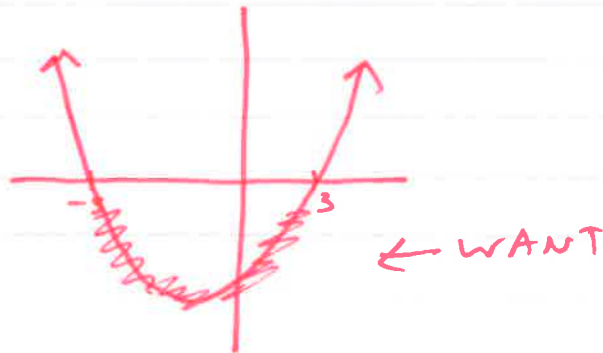
vertex $(2, -36)$

11e)

$$-x^2 - 2x + 15 \geq 0$$

$$x^2 + 2x - 15 \leq 0$$

$$(x+5)(x-3) \leq 0$$



$$-5 \leq x \leq 3 \quad \checkmark \checkmark$$

f)

$$\frac{\sin x}{\cos x} + \frac{\cos x}{\sin x} = \operatorname{cosec} x \sec x$$

$$\text{LHS} = \frac{\sin x}{\cos x} + \frac{\cos x}{\sin x}$$

$$= \frac{\sin x \times \sin x}{\sin x \cos x} + \frac{\cos x \times \cos x}{\sin x \cos x}$$

$$= \frac{\sin^2 x + \cos^2 x}{\sin x \cos x} \quad \checkmark$$

$$= \frac{1}{\sin x \cos x}$$

$$= \frac{1}{\sin x} \times \frac{1}{\cos x}$$

$$= \operatorname{cosec} x \sec x$$

$$= \text{RHS} \quad \checkmark$$

11g) $l_1: y = 2x + 4$ concurrent $\Rightarrow$ all intersect at same point *

$l_2: -(y = x + 14)$

$$0 = x - 10$$

$$x = 10$$

when $x = 10$, $l_1: y = 2 \times 10 + 4$

$$y = 24$$

✓ both x and y

$x = 10, y = 24$ in $l_3: y = 3x - 6$

$$LHS = 3x - 6$$

$$= 3 \times 10 - 6$$

$$= 30 - 6$$

$$= 24$$

$$RHS = y$$

$$= 24$$

$LHS = RHS$ so l_1, l_2, l_3 concurrent ✓

Question 12

a) $x = \frac{1 + 4x}{x}$

$$x^2 = 1 + 4x$$

$$x^2 - 4x - 1 = 0$$

$$(x-2)^2 - 4 - 1 = 0$$

$$(x-2)^2 - 5 = 0$$

$$x - 2 = \sqrt{5}$$

$$x = 2 + \sqrt{5}$$

or $x - 2 = -\sqrt{5}$

$$x = 2 - \sqrt{5}$$

✓ both

$$12b) \quad \frac{2}{\sqrt{3}-\sqrt{2}} = m(\sqrt{2}+\sqrt{3})$$

$\frac{2}{(\sqrt{3}-\sqrt{2})(\sqrt{2}+\sqrt{3})} = m \quad \checkmark$ $m = \frac{2}{3-2}$ $m = \frac{2}{1}$ $m = 2 \quad \checkmark$	O R	$\frac{2}{\sqrt{3}-\sqrt{2}} \times \frac{\sqrt{3}+\sqrt{2}}{\sqrt{3}+\sqrt{2}} \quad \checkmark$ $= \frac{2\sqrt{3}+\sqrt{2}}{3-2}$ $= 2(\sqrt{2}+\sqrt{3})$ so $m=2$ by inspection $\checkmark$
--	-----	--

c i)

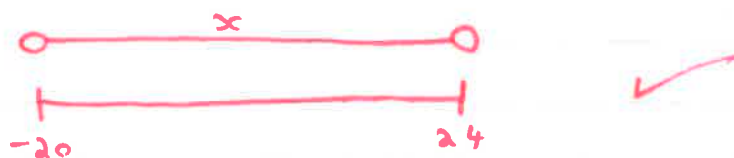
$$|x-2| < 22$$

$x-2 < 22$
 $x < 24 \quad \checkmark$

$-(x-2) < 22$
 $x-2 > -22$
 $x > -20 \quad \checkmark$

so $-20 < x < 24$

ii)



di)

$$x^3 - 7x^2 + 10x = x(x^2 - 7x + 10) \quad \checkmark$$

$$= x(x-2)(x-5) \quad \checkmark$$

ii)

$$0 = x(x-2)(x-5)$$

$$x=0 \quad \text{or} \quad x-2=0 \quad \text{or} \quad x-5=0$$

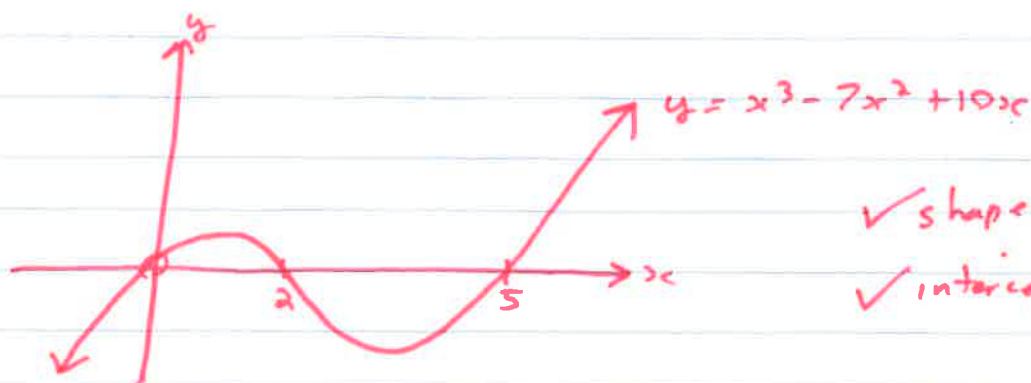
$$x=0 \quad \text{or} \quad x=2 \quad \text{or} \quad x=5 \quad \checkmark \text{ all 3}$$

12 d iii)

when $x=10$, $y=400$

so

(not $\sim$)



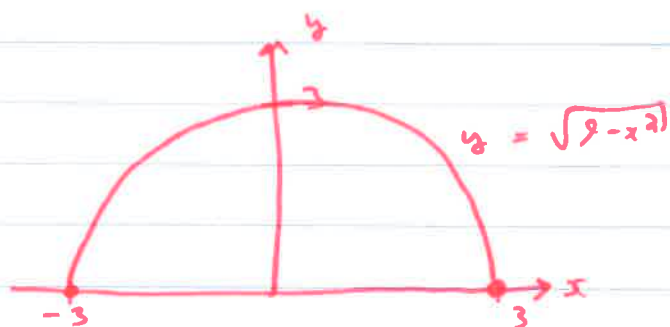
✓ shape

✓ intercepts

Question 13

$$\begin{aligned}
 a) \quad \frac{6x-30}{3x^2-75} \div \frac{1}{4x+20} &= \frac{6(x-5)}{3(x^2-25)} \times \frac{4x+20}{1} \\
 &= \frac{6(x-5)}{3(x+5)(x-5)} \times \frac{4(x+5)}{1} \quad \checkmark \\
 &= \frac{6 \times 4}{3 \times 1} \\
 &= \frac{24}{3} \\
 &= 8 \quad \checkmark
 \end{aligned}$$

b i)



✓ shape

✓ intercepts

ii) values on x-axis, so

Domain: all real values of x , where $-3 \leq x \leq 3$ ✓

iii) values on y-axis, so

Range: all real values of y , where $0 \leq y \leq 3$ ✓

13c)



use sine rule

$$\frac{\sin \theta}{4\sqrt{2}} = \frac{\sin 30}{4}$$



$$\sin \theta = \frac{4\sqrt{2} \sin 30}{4}$$

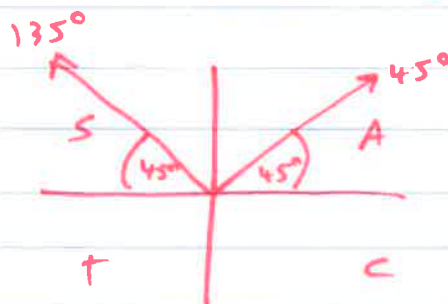
$$\sin \theta = \frac{4\sqrt{2} \times \frac{1}{2}}{4}$$

$$= \frac{2\sqrt{2}}{4}$$

$$\sin \theta = \frac{\sqrt{2}}{2}$$

$$\text{related } \theta = \sin^{-1}\left(\frac{\sqrt{2}}{2}\right)$$

$$= 45^\circ$$



so $\theta = 45^\circ$ or 135°

d) $M = \left(\frac{2+8}{2}, \frac{10+0}{2}\right)$

$$= (5, 5)$$



gradient $m = \frac{10-0}{2-8}$

$$= -\frac{10}{6}$$

$$= -\frac{5}{3}$$

perpendicular gradient $= \frac{3}{5}$



$$y - 5 = \frac{3}{5}(x - 5)$$

$$y - 5 = \frac{3}{5}x - 3$$

$$y = \frac{3}{5}x + 2$$

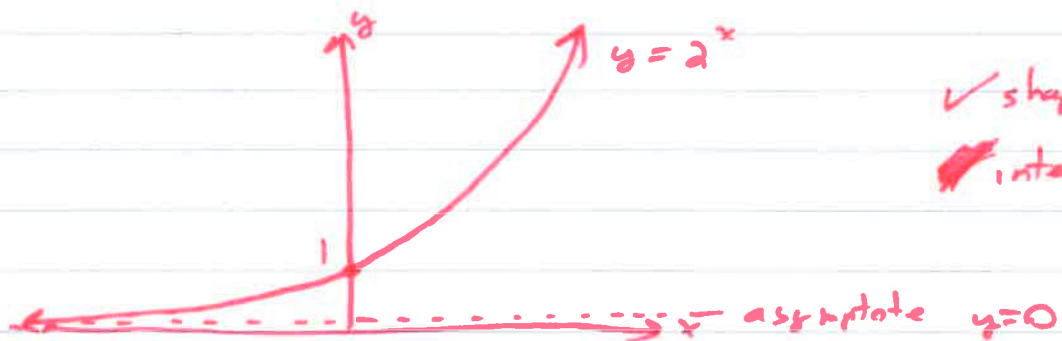


Question 14

a) $f(x) = x(x^2 - 4)$
 $f(-x) = -x((-x)^2 - 4)$
 $= -x(x^2 - 4)$
 $= -f(x)$

so ODD ✓

bi)



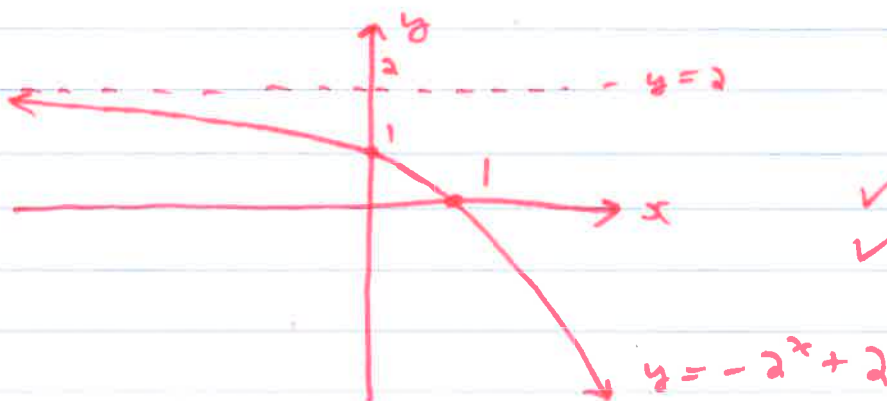
✓ shape and
 ✓ intercept

iii)

$y = -2^x + 2$
 first reflect in x-axis

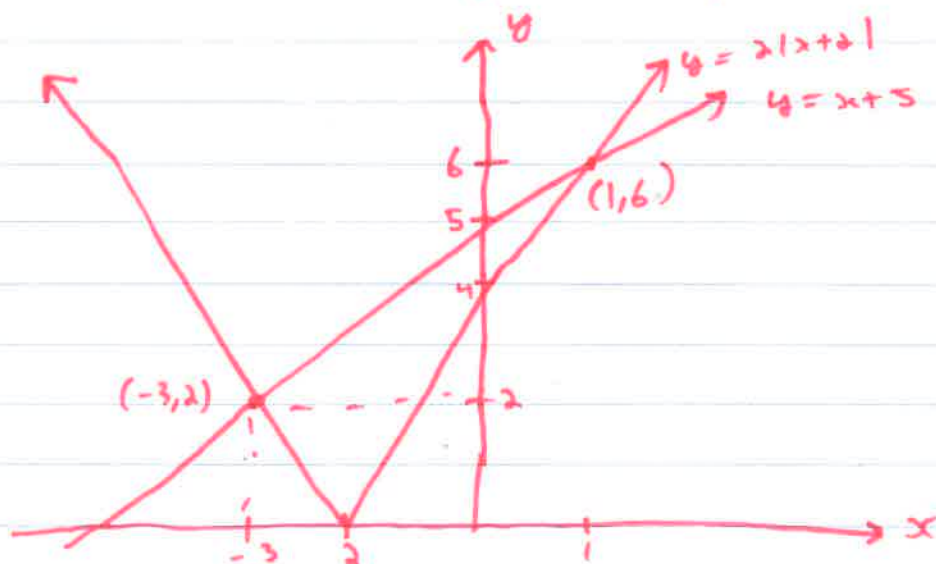


then shift up by 2:



✓ shape
 ✓ x-intercept

ci)



✓ both shape

14ciii)

$$2|x+2| = x+5$$

$$2(x+2) = x+5$$

$$2x+4 = x+5$$

$$x = 1$$

$$y = 6$$

$$-2(x+2) = x+5$$

$$-2x-4 = x+5$$

$$-9 = 3x$$

$$x = -3$$

$$y = 2$$

$$(1, 6) \text{ and } (-3, 2)$$

iii) where $y = 2|x+2|$ is below $y = x+5$

$$-3 < x < 1$$

d)

$$\operatorname{cosec} x - \sin x = \cot x \cos x$$

$$\text{L.H.S.} = \operatorname{cosec} x - \sin x$$

$$= \frac{1}{\sin x} - \frac{\sin x}{1}$$

$$= \frac{1}{\sin x} - \frac{\sin^2 x}{\sin x}$$

$$= \frac{1 - \sin^2 x}{\sin x}$$

$$= \frac{\cos^2 x}{\sin x}$$

$$= \frac{\cos x}{\sin x} \times \cos x$$

$$= \cot x \cos x$$

e)

$$p^2 = \left(\frac{1}{\sqrt{3}} + \frac{1}{\sqrt{5}} \right)^2$$

$$= \frac{1}{3} + \frac{1}{5} + \frac{2}{\sqrt{15}}$$

$$= \frac{8}{15} + \frac{2}{\sqrt{15}}$$

$$= \frac{8}{15} + \frac{2\sqrt{15}}{15}$$

$$= \frac{8 + 2\sqrt{15}}{15}$$

$$\sqrt{15} = ap^2 + b$$

PRO
→

14 e continued...)

$$\text{so } \sqrt{15} = ap^2 + b = \frac{8a}{15} + \frac{2\sqrt{15}a}{15} + b$$

compare irrational numbers : $\sqrt{15} = \frac{2\sqrt{15}}{15} \times a$

$$a = \frac{15}{2} \quad \checkmark$$

compare rational numbers : $b + \frac{8a}{15} = 0$

$$b + 4 = 0$$

$$b = -4 \quad \checkmark$$

$$\text{so } a = \frac{15}{2} \text{ and } b = -4.$$